

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

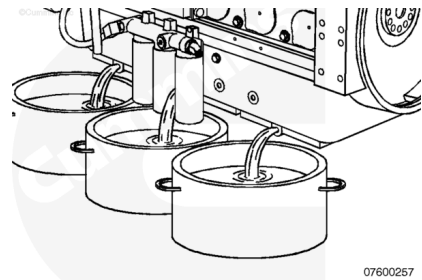


Disconnect the batteries.

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⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



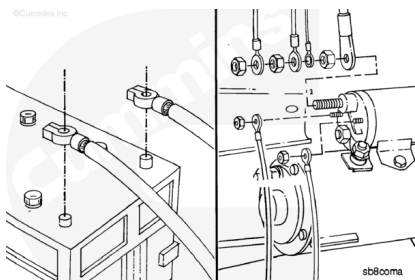
⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>).

Remove

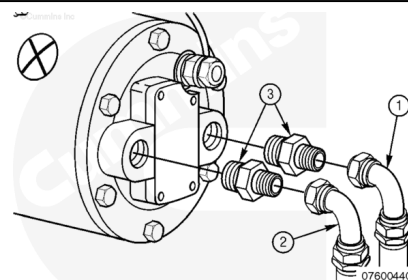
Disconnect the electrical connections from the starting motor



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Remove the suction hose (1) from the hose fitting (3) on the back of the integral prelube starter. Disconnect the other end of the suction hose from the oil pump.

Remove the supply hose (2) from the hose fitting (3) on the back of the integral prelube starter. Disconnect the other end of the supply hose from the oil filter head.



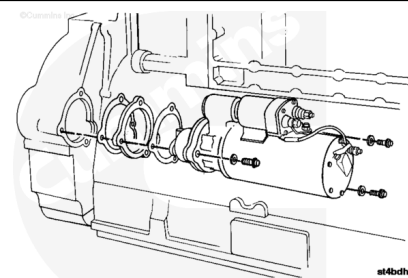
⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get personal assistance to lift this assembly.

Remove the starting motor capscrews, the starter, the spacers, and the gaskets.

Discard the gaskets.

Note : Not all engines contain spacers and gaskets.



Inspect for Reuse

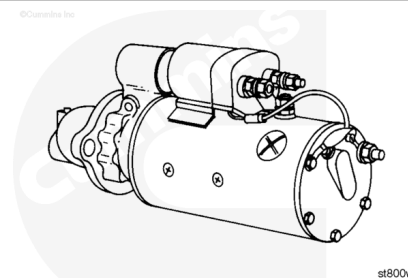
⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

Clean the exterior of the motor with steam.

Inspect the gear, shaft, and bushing for wear or damage.

Inspect flywheel ring gear for wear or damage.



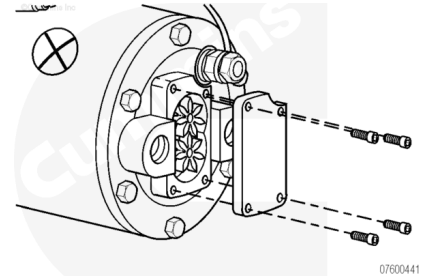
Lubricate the bushing with clean engine oil.

Note : A pipe plug **must** be removed to lubricate the bushing on some starter motors.

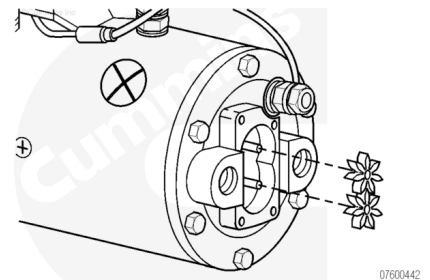
Disassemble

Remove the four end cover capscrews.

Remove the end cover.



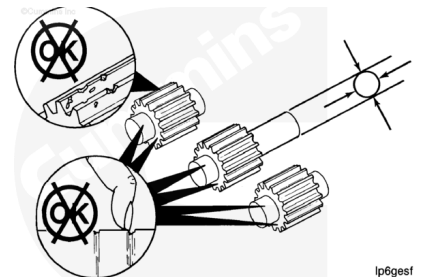
Slide the two pumping gears off the shafts.



Inspect for Reuse

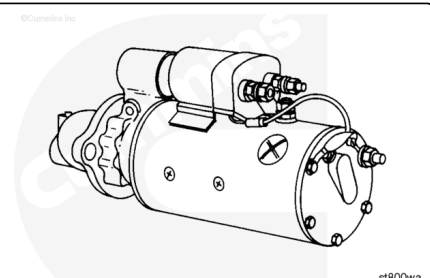
Inspect the gears for cracks, chipped or broken teeth.

If the gears are damaged, the gears **must** be replaced.



Inspect the gear, shaft, and bushing for wear or damage.

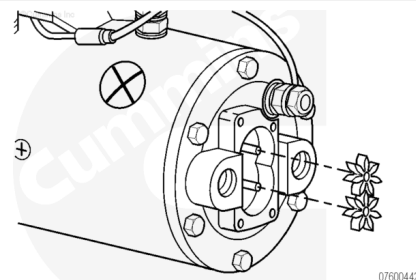
Note : A pipe plug **must** be removed to lubricate the bushing on some starter motors.



Assemble

Lubricate the gears with clean engine oil.

Slide the gears onto the shafts.

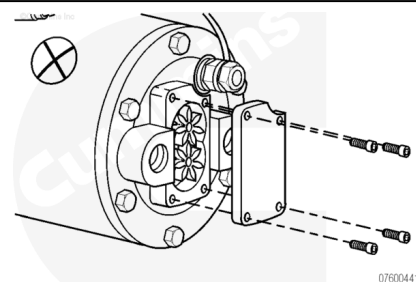


Install the cover plate.

Install the four cover plate capscrews.

Tighten the capscrews.

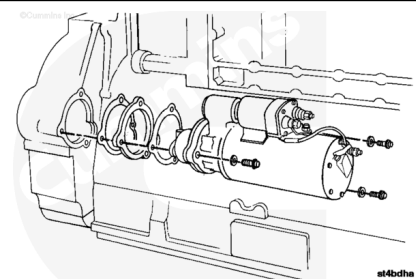
Torque Value: 18 n•m [159 in-lb]



Install

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get personal assistance to lift this assembly.



Note : Not all engines use spacers.

Note : The wet type of flywheel housing requires gaskets for the starting motor.

Install any spacer or gaskets.

Install the starting motor and capscrews.

Torque Value:

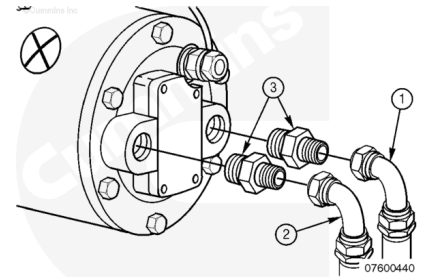
With Cast-Iron Flywheel Housing

1. 180 n•m [133 ft-lb]

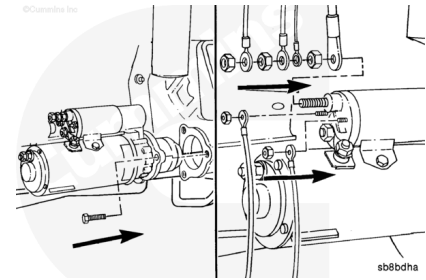
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Install the suction hose (1) to the fittings (3) on the back of the integral prelube starter. Install the other end of the suction hose to the suction hose of the oil pump.

Install the supply hose (2) to the fittings (3) on the back of the prelube starter. Install the other end of the supply hose to the oil filter head.



Install the electrical connections to the starting motor.



Finishing Steps

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Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Connect the batteries.



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Fill the engine with lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>).

